MODEL IDENTIFICATION AND DATA ANALYSIS – Part 2 (Automation&Control Engineering), A.Y. 2021/2022

Prof. Sergio M. Savaresi, Prof. Simone Formentin, Prof. G. Panzani – **February 3rd, 2022**

Surname	**Name**	**University ID Number**	**Signature**
....	....	....	....

===

- Write the solutions (including procedures and intermediate steps) in the blank areas (use the back of the page, if needed)
- The number of pages is 4. Additional papers will not be considered.
- Clarity, order and precision are strongly considered for the final evaluation.

===

1. [Identification]

Consider a model belonging to the following class

$$\mathcal{M}: \quad y(t) = a \cdot y(t-1) + e(t)$$

where $e(t) \sim WN(m_e, \lambda^2)$ and $a \in \mathbb{R}$ is a model parameter. Additionally, consider the following measured samples:

$$y(1) = 0.5, \qquad y(2) = 1.9, \qquad y(3) = 3.4, \qquad y(4) = 1.7$$

Answer the following questions.

a. Discuss for which values of the parameter a the modelled process is stationary
b. Compute the time domain expression of the 1-step ahead output predictor
c. Considering $m_e = 0$ identify, using the available data, the value of the parameter a.
d. Repeat the identification of the parameter a considering $m_e = 1$. Which of the two identified model should be considered the most reliable description of the process that generated the measured data?

Solutions

a. The process is an AR(1) that can be written in the following transfer function form

$$y(t) = W(z)e(t) = \frac{1}{1 - az^{-1}} e(t) \qquad \text{with} \quad e(t) \sim WN(m_e, \lambda^2)$$

In order to generate a stationary process, the transfer function must be asymptotically stable, which happens when $|a| < 1$.

b. When writing the 1-step ahead prediction, it's important to notice that the process has a non-zero mean value $\left(m_y = \frac{1}{1-a} m_e\right)$, resulting from the non-zero mean noise that feeds the system. Thus, firstly the predictor for the de-biased process $\tilde{y} = y(t) - m_y$ is computed:

$$\hat{\tilde{y}}(t+1|t) = a \cdot \tilde{y}(t)$$

Then, the predictor for the process output is computed

$$\hat{y}(t+1|t) = \hat{\tilde{y}}(t+1|t) + m_y = a \cdot \left(y(t) - \frac{1}{1-a} m_e\right) + \frac{1}{1-a} m_e = a \cdot y(t) + m_e$$

c. In order to identify the parameter a the sampled prediction error variance is minimized:

$$J_1(a) = \frac{1}{3} \sum_{t=2}^{t=4} \left(y(t+1) - \hat{y}(t+1|t)\right)^2 = \frac{1}{3}(18.06 - 26.38 \cdot a + 15.42 \cdot a^2)$$

which is a quadratic function of a. Its minimum corresponds to the following value of a:

$$a = \frac{26.38}{30.84} = 0.85$$

d. In order to identify the parameter a when $m_e = 1$ the very same procedure is done (obviously, the predicted output values change, according to the previous predictor expression). It results that:

$$J_2(a) = \frac{1}{3}\sum_{t=2}^{t=4}\big(y(t+1) - \hat{y}(t+1|t)\big)^2 = \frac{1}{3}(7.06 - 14.78 \cdot a + 15.42 \cdot a^2)$$

which is again a quadratic function of a. Its minimum corresponds to the following value of a:

$$a = \frac{14.78}{30.84} = 0.47$$

In order to compare the two identified models, the corresponding optimal cost function value can be compared, as in both cases it represents the sampled prediction error variance. We obtain

$$J_1(a = 0.85) = 2.28$$
$$J_2(a = 0.47) = 1.17$$

Thus the second model, having a smaller variance, is more reliable in describing the process that generated the measured data.

2. [Kalman filter]

Consider the following dynamic system

$$\begin{cases} x(t+1) = \dfrac{1}{2}x(t) + v_1(t) + c \cdot v_2(t) \\ y(t) = v_1(t) \end{cases}$$

where $v_1(t) \sim WN(0,1), v_2(t) \sim WN(0,1)$ and $v_1(t) \perp v_2(t)$.

Answer the following questions.

a. Compute the Kalman filter framework matrices F, H, V_1, V_2, V_{12}. (Properly describe why and how they are computed – do not just write the result!).
b. From now on, consider $c = 0$. Is the variance of the asymptotic 1-step ahead state prediction error limited? If so, compute its value.
c. Write the equations of the asymptotic 1-step ahead Kalman predictor.
d. Compute the variance of the asymptotic 1-step ahead output prediction error and compare its value with the state prediction error one. Comment the obtained result, in the light of the system model and the obtained Kalman predictor.

Solutions

a. In the Kalman framework we have that:

F is the state matrix, the one that multiples the state $x(t)$ in the state equations. Inspecting the dynamic system equations, we have that

$$\boldsymbol{F} = \left[\frac{1}{2}\right].$$

H is the output matrix, the one that multiples the state $x(t)$ in the output equations. Inspecting the dynamic system equations, we have that

$$\boldsymbol{H} = [\boldsymbol{0}].$$

V_1 is the variance of the noise that feeds the state equations. Inspecting the dynamic system equations, we have that:

$$\boldsymbol{V_1} = Var[v_1 + c \cdot v_2] = Var[v_1] + Var[c \cdot v_2] = 1 + c^2 \cdot 1 = \boldsymbol{1 + c^2}$$

V_2 is the variance of the noise that feeds the state equations). Inspecting the dynamic system equations, we have that

$$\boldsymbol{V_2} = Var[v_1] = \boldsymbol{1}.$$

V_{12} is the correlation between the noise that feeds the state equations and the one that feeds the output equation. Inspecting the dynamic system equations, we have that

$$\boldsymbol{V_{12}} = \mathbb{E}[(v_1 + c \cdot v_2) \cdot v_1] = \mathbb{E}[v_1^2] + \mathbb{E}[c \cdot v_2 \cdot v_1] = 1 + 0 = \boldsymbol{1}$$

b. Since $V_{12} \neq 0$ we cannot take advantage of the convergence theorems. However, since the system is scalar, we can study the DRE steady-state solutions (if any) stability via the graphical method.

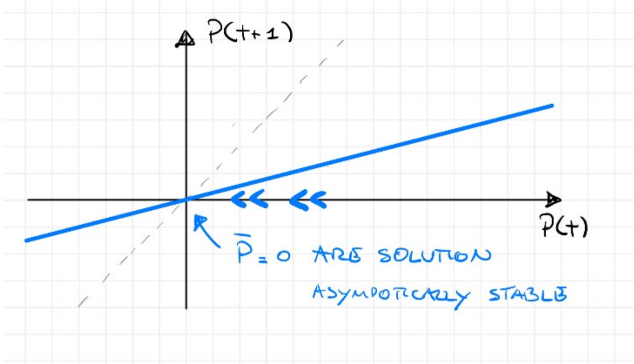
First, the DRE is computed:

$$P(t+1) = \frac{1}{4}P(t) + 1 - \frac{1^2}{1} \rightarrow P(t+1) = \frac{1}{4}P(t)$$

Any ARE solution should satisfy

$$\bar{P} = \frac{1}{4}\bar{P} \rightarrow \bar{P} = 0$$

So, there is only one steady state solution.
The graphical method (see the following figure) allows to certify the stability of the found solution.



Thus, one can conclude that not only the asymptotic 1-step ahead state prediction error is limited but is also equal to zero. This means that we can predict the system state exactly. Inspecting the system equations this does not surprise us: in fact, the measured output, when $c = 0$ is exactly the noise that feeds the state equation. Thus, there is no uncertainty in the state equation and its evolution can be computed deterministically.

c. The Kalman predictor asymptotic gain is:

$$\bar{K} = 1$$

and the resulting predictor

$$\begin{cases} \hat{x}(t+1|t) = \frac{1}{2}\hat{x}(t|t-1) + y(t) \\ \hat{y}(t|t-1) = 0 \end{cases}$$

Its expression confirms what previously discussed. In fact, the state equation is nothing but a "copy" of the original system, fed by the measured output, which replaces the process noise.

d. The asymptotic 1-step ahead output prediction variance is:

e.

$$\varepsilon_y(t+1|t) = y(t) - \hat{y}(t|t-1) = y(t) \rightarrow Var[\varepsilon_y(t+1|t)] = Var[y(t)] = Var[v_1(t)] = 1.$$

whereas, as previously recalled:

$$\varepsilon_x(t+1|t) = P(t) = 0$$

So, we have a perfect prediction of the system state (as previously explained), but not of the system output. As the system equations suggest, since the measured output is a white noise, it cannot be, by definition, predicted. This is consistent with the Kalman predictor output equation, which features a null value. For this reason, the output prediction error variance is equal to the output noise variance, $Var[v_1] = 1$.

3. [ARMA/ARMAX models]

a) Describe the all-pass filer structure

b) Discuss its properties

See lecture notes

4. [ARMA/ARMAX models]

Find (proof requested) the formula for the optimal predictor from data of an ARMAX model.

See lecture notes